MNIST Digit Classifier

A small PyTorch CNN that classifies handwritten digits (0-9) from the MNIST dataset.

What it does

- Downloads MNIST automatically (60k train / 10k test images, 28x28 grayscale)
- Trains a 2-layer convolutional network
- Prints train/test accuracy each epoch - expect ~99% test accuracy after 5 epochs
- Saves trained weights to `mnist_cnn.pt`

Run it

```
pip install -r requirements.txt
python train.py
```

GPU is used automatically if available, otherwise it trains on CPU (a few minutes).

Things to try next

- **Predict your own digits**: load `mnist_cnn.pt` and run inference on a hand-drawn digit image
- **Tune it**: change learning rate, batch size, dropout, or add a third conv layer and watch accuracy move
- **Visualize**: plot the first conv layer's filters, or use a confusion matrix to see which digits get mixed up